

Let's see whether we have languages we cannot recognize by an NFA/DFA.

consider the following languages.

$A = \{w : w \text{ has more } 0\text{'s than } 1\text{'s}\}$

$B = \{a^n b^n \mid n \geq 0\}$

$C = \{w \mid w \text{ has equal number of } 0\text{'s and } 1\text{'s}\}$

When you consider language A,
If you want to build a DFA
that recognizes A, the machine
needs to keep track of # of
0's or 1's, and depending on the
string this value could be different.

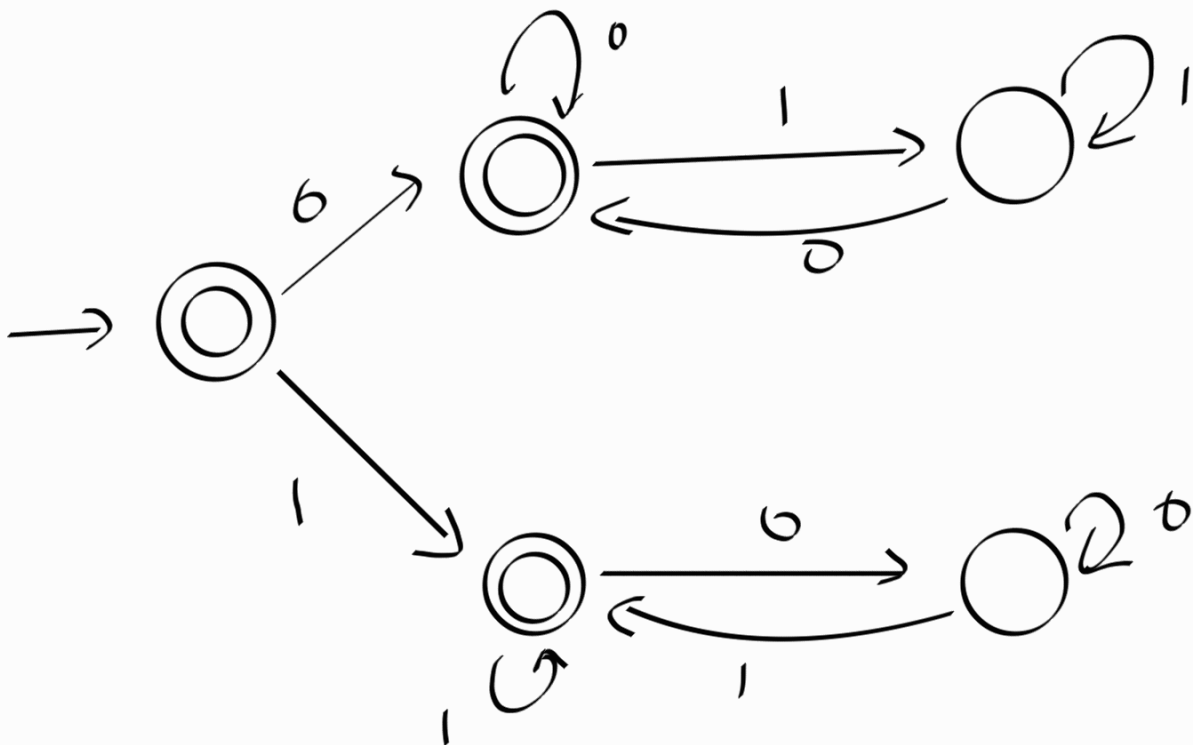
Therefore, a machine with limited
amount of memory cannot do it.

However, just because a language appears to require unbounded memory does not mean that the language is not regular.

$$C = \{w \mid w \text{ has an equal \# of } 0's \text{ and } 1's\}$$

$$D = \{ w \mid w \text{ has equal \# of occurrences of 01 and 10} \}$$

Σ 0 00000 1 1111 ✓
✓ ✓ ✓ ✓
00001100 0000110001



Pumping Lemma

- Basically, this theorem states that regular languages have a special property.
- If we can show that a language does not have this property, then the language is not regular.

The basic idea of this property is that all strings in a regular language can be "pumped", if they are at least as long as a certain special value, called the pumping length.

Pumping Lemma

If A is a regular language, then there is a number p (the pumping length) where if s is any string in A of length at least p , then s may be divided into three pieces, $s = xyz$, satisfying the following conditions.

1. for each $i \geq 0$, $xy^iz \in A$
 2. $|y| > 0$, and
 3. $|xy| \leq p$
- $|s|$ represents the length of the string
 - y^i means i copies of string y .

$$A \text{ is regular} \Rightarrow \left[\exists p \in \mathbb{N} : s \in A \wedge |s| \geq p \Rightarrow s = xyz \right]$$

1.

2.

3.

Proof idea:

Assume A is regular.

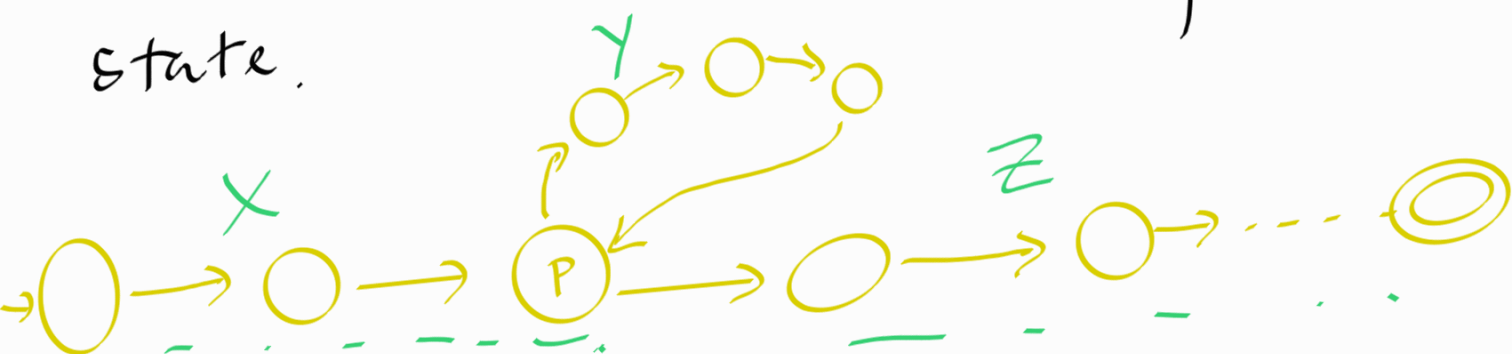
Then A can be recognized by some DFA, let's call it M .

We pick p to be the # of states in DFA M (that recognize language A).

Then pick any string s in A of length at least p .

* If no such string exist, then the claim is vacuously true.

Since the size of s is larger than ^{or equal to} p , then in its execution path to an accepting state, we must see a repeated state.



$$\underline{s = xyz}$$

$$xy^iz \in A$$

Since this path leads to an accepting state, this path should accept strings, $xz, xyx, xy^2z, xy^3z, \dots, xy^iz, \dots$

Question! Can we use pumping lemma to show that a particular language is regular?

We do not use pumping lemma to show a language is regular.

In fact, we cannot do this as this lemma is not bidirectional.

Just because a language follows pumping lemma it does not mean it is regular.

Ex: $L = \{ab^n c^n; n \geq 0\} \cup \{a^k w; k \neq 1 \text{ and } w \in \Sigma^* \text{ does not start with } a\}$
 $\Sigma = \{a, b, c\}$

← non-regular but follows pumping lemma.

But we can use this to show a language is not regular.